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Effects of Trang Phuc Linh Plus-Food Supplement on Irritable Bowel Syndrome Induced by Mustard Oil

Thong T. Nguyen,¹ Duong T. Dau,¹ Dung C. Nguyen,¹ Huong T.T.G. Nguyen,¹
Phuong T.B. Ngo,² Thai Q. Nguyen,² Bo Han,³ and Ba X. Hoang³

¹Department of Pharmacology, Hanoi Medical University, Hanoi, Vietnam.

²Thai Minh Pharmaceutical JSC, Hanoi, Vietnam.

³Department of Surgery, Keck School of Medicine, University of Southern California, Los Angeles, California.

ABSTRACT Trang phuc linh plus (TPLP) is a food supplement product derived from dried extracts of herbal agents *Atractylodes macrocephala*, *Poria cocos*, *Paeonia lactiflora*, *Phellodendron amurense*, and added *Lactobacillus fermentum* lysate (ImmuneGamma®) and 5-hydroxytryptophan. TPLP is a functional food used as adjunctive treatment for treating irritable bowel syndrome (IBS). However the biological effect and its mechanism of action in IBS have not been elucidated. In this study, we aimed to determine the pharmacological activities and mode of action of TPLP on IBS animal models. Mice were given a single administration of 5% mustard oil (MO) intracolonic. Acute colitis induction by MO resulted in later development of an IBS-like accelerated upper gastrointestinal transit in mice. Mice were treated with different doses of TPLP and controls. Results showed that TPLP at the dose of 654 mg/kg/day given orally significantly decreased intestinal motility (IM) compared with the control animals. The effect was similar to Duspatalin (80 mg/kg/day) (Mebeverine Hydrochloride, an antispasmodic that helps to relieve the pain and discomfort associated with gastrointestinal spasms). Increased TPLP dose (1962 mg/kg/day) had a better effect on relief of IM than Duspatalin (80 mg/kg/day). TPLP also reduced peristalsis frequency and decreased fluid volume and electrolytes excretion in intestine tested in *ex vivo* models. Overall, TPLP may be an effective nutraceutical supplement for IBS.

KEYWORDS: • animal models • mustard oil • trang phuc linh plus • intestinal motility • irritable bowel syndrome

INTRODUCTION

IRRITABLE BOWEL SYNDROME (IBS), affecting ~10–15% of populations, is one of the most common disorders of the gastrointestinal tract.^{1,2} Symptoms of IBS include abdominal pain, cramping, change in bowel habits, and disturbance of defecation.^{1,2}

IBS is not only highly prevalent, but it also induces considerable human suffering and a major economic burden on society.² Despite the increasing efforts of pathophysiological and therapeutic research in the last 20 years, IBS remains a therapeutic challenge and poorly understood disorder.^{3,4}

The use of nutritional and food supplements to manage symptoms of IBS becomes in trend due to the lack of safe and effective drugs for long-term application.^{5,6} Trang phuc linh plus (TPLP) is a food supplement product derived from herbal ingredients including *Atractylodes macrocephala*, *Poria cocos*, *Paeonia lactiflora* extractum, *Phellodendron amurense*, probiotic (ImmuneGamma®), and 5-hydroxytryptophan (5-HTP). TPLP is approved by Vietnamese Health Ministry as a food

supplement to support the treatment of IBS. In this study we evaluated the effects of TPLP on IBS treatment in experimental animal models.

MATERIALS AND METHODS

Animals

Healthy Swiss mice, either sexes, weighing 18–22 g supplied by the National Institute of Hygiene and Epidemiology. Both New Zealand white rabbits, either sexes, weighing 2.2–2.5 kg, and frogs, weighing 200–220 g, either sexes, were supplied by Bavi Center (Hanoi, Vietnam). Animals were acclimated 5–7 days before experiments to adapt to the breeding conditions. Before and during the study, animals were allowed to eat food and drink water freely.

The experiments were carried out in Department of Pharmacology, Hanoi Medical University, Vietnam. Animal research protocols were approved by the Institutional Animal Care and Use Committee (IACUC) of Hanoi Medical University.

Drugs and chemicals

TPLP formulation. One TPLP film-coated tablet contains 200 mg aqueous extract of *rhizoma Atractylodis macrocephalae* (*Atractylodes*), 50 mg aqueous extract of *Poria*

Manuscript received 16 August 2016. Revision accepted 4 February 2017.

Address correspondence to: Ba X. Hoang, MD, PhD, Nimmi-Cordoba Tissue Engineering and Drug Discovery Laboratory, Department of Surgery, Keck School of Medicine, University of Southern California, Los Angeles, CA 90089, E-mail: baxuanho@usc.edu

cocos, 50 mg aqueous extract of *radix Paeoniae lactiflorae* (as known as Chinese peony), 50 mg aqueous extract of *cortex Phellodendri*, 100 mg of ImmuneGamma (muramyl peptides extracted from *Lactobacillus fermentum*), 3 mg of 5-HTP. All plant ingredients were extracted and analyzed by standard TLC methods listed in Vietnamese Pharmacopoeia IV (2009). TPLP is manufactured by International Medical Consulting Co., Ltd, Vietnam.

Duspatalin (Mebeverine[®], Solvay Pharmaceuticals B.V), mustard oil (MO) (Allyl isothiocyanate 94%, Acros Organics N.V, Belgium), Tyrode's solution, activated carbon (AC), carboxymethyl cellulose (CMC), ethanol, and other chemicals were purchased from Vinamask Co., Ltd.

The rationale for using antispasmodic agents is to attenuate the heightened baseline and postprandial contractility seen in patients with IBS (particularly when diarrhoea predominant). Mebeverine is generally well tolerated and can be used as needed basis (before meals) to improve symptoms.⁷

Charcoal meal test for intestinal motility

In vivo effect of TPLP on intestinal motility (IM) was evaluated based on the movement of AC following the method described.^{8,9} Specifically, total 72 mice were randomly divided into 4 lots with 2 time points ($n=9$). Mice were fasted but allowed to drink water freely 20 h before experiments. Then mice were fed with either drug or water through gavage for 5 days once daily. Four treatment lots were included. Lots 1: pure water, Lot 2: Duspatalin 80 mg/kg/day, Lot 3: TPLP dose (654 mg/kg/day), Lot 4: TPLP dose (1962 mg/kg/day). One hour after day 5 treatment, all mice were given 0.2 mL AC gel (10 g of AC suspended in 100 mL 3% CMC) through gavage. In each lot, total of nine animals were sacrificed at either 20 or 40 min post AC feeding to retrieve bowel. The distance of charcoal transited from pylorus to passage black streak (color of AC) was measured. The IM was calculated as percentage of the intestinal length with AC per the intestinal length from pylorus to cecum.

Peristalsis test on isolated intestinal segments

The peristalsis study was carried out on rabbits' isolated intestinal segments following the method described as Magnus Technique.¹⁰ Briefly, rabbits, weighing around 3 kg, were anesthetized by thiopental. Each of 2-cm intestinal segments was excised through an abdominal midline incision. After cleaning with an intraluminal flush of Tyrode solution, intestinal segments were suspended in an organ bath with aeration and controlled temperature (37°C) for 30 min to stabilize the tissue.

Tyrode's solution is a solution that is roughly isotonic with interstitial fluid and used in physiological experiments and tissues culture. It resembles lactated Ringer's solution, but contains magnesium, glucose as an energy source and uses bicarbonate and phosphate as a buffer instead of lactate.

In this experiment, Tyrode's solution was used to maintain the normal peristalsis of isolated intestinal segments after removing from the rabbits.

For experiment, four TPLP treatment concentrations were prepared in Tyrode solution, for example, Lot 1: 327 mg TPLP/100 mL Tyrode's solution; Lot 2: 654 mg TPLP/100 mL Tyrode's solution; Lot 3: 1308 mg TPLP/100 mL Tyrode's solution; Lot 4: 1962 mg TPLP/100 mL Tyrode's solution. Total six rabbit intestinal segments were used in each condition ($n=6$). Spontaneous contractions of the intestinal segments were recorded before and after TPLP additions.

Absorption of water and electrolytes from intestine

The experiment was carried out in frogs following the method of Holschneider *et al.*¹¹ Briefly, after breaking marrow and evisceration, small intestine was pulled out while maintaining mesenteric and intestinal blood vessels intact. Roughly 3 cm down from the pylorus, at which intestine has uniform cross section, a closed intestine section was created by suture ligation with 2 cm in length. A volume of 1 mL of either drug or water was applied into the ligated segments by 27-G needle injection. Thirty-two frogs were randomly divided into four lots, with eight frogs in each treatment ($n=8$). Test TPLP solution was prepared in pure water as follows: Lot 1, 1 mL of pure water; Lot 2, 1 mL of 0.08% Duspatalin; Lot 3, 1 mL of 0.654% TPLP; and Lot 4, 1 mL of 1.962% TPLP. After injection, intestine was then put back into frog abdominal cavity and abdominal wall was closed and animals were kept at room temperature. After 60 min, fluid in the ligated section was aspirated by needle suction. The volume of the fluid was measured and the concentrations of sodium, chloride, and potassium in intestinal fluid were analyzed by Atomic Absorption Spectrometer at Laboratory Department of Vietnam-Cuba Hospital.

Evaluate the effect of TPLP on IBS induced by MO

Acute colitis was induced by MO to achieve the development of IBS-like gastrointestinal disorders in mice. IBS model was carried out following the method established by Kimball *et al.*⁹ Experiment was carried out to evaluate the TPLP effect on IBS therapeutics. As illustrated in Figure 1, total of 48 mice were divided into 6 groups ($n=8$). Mice were anesthetized by IP injection of thiopental (50 mg/kg). To induce IBS, total of 100 μ L of 5% MO solution (diluted in 30% ethanol) was administered intracolonic to a depth of 4 cm via a catheter. Two control groups were administered with either (A) saline or (B) 30% ethanol vehicle. In group C, animals were carefully monitored of IBS symptoms such as diarrhea (volume and frequency), body weight change, and activity for 7 days. The established IBS animals were further divided into four subgroups, C-0: non-treated; C-1: treated with 80 mg/kg/day of Duspatalin as positive control; C-2: low dose of TPLP (654 mg/kg/day); and C-3: high dose TPLP (1962 mg/kg/day).

Mouse IBS model was considered successfully established when the colon's lesions (inflammatory cell infiltration, epithelial lesion, and lesion in the smooth muscle) recovered but the symptoms (diarrhea and increased gut motility) still remained.

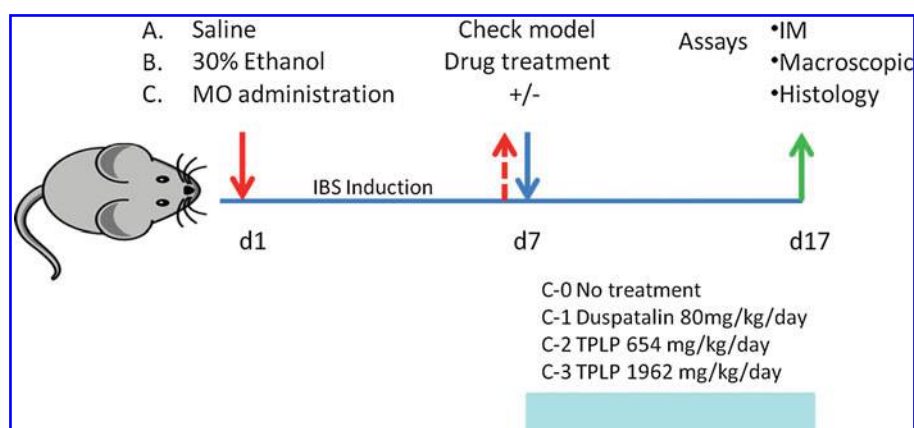


FIG. 1. Experimental schedule for evaluating the treatment of TPLP in IBS induced by MO on animals. IBS, irritable bowel syndrome; MO, mustard oil; TPLP, trang phuc linh plus.

At the termination of experiments, macroscopic and microscopic examinations of the colons were carried out and scored. Colons were resected, examined for signs of inflammation, and weighed after removing fecal contents (which were examined for signs of diarrhea). Colon histological examination was carried out to determine the lesions of epithelium, destruction of mucosal architecture, and other lesions.

Colon macroscopic and microscopic scores

Colon histology (microscopic): the colon was dissected from 1 to 4 cm from the anus to evaluate histological lesions (including epithelium injury, cell infiltration, and damage or alteration of smooth muscle). The score ranged from 0 to 4, as defined in Tables 1 and 2.

Statistical analysis

Data represent mean $\pm$ standard deviation. Experimental groups were analyzed for significance of differences between the means of treatment groups and control groups by ANOVA. The difference between two groups is judged to be statistically significant when $P \leq .05$.

RESULTS

In vivo effect of TPLP on the movement of AC

As shown in Table 3, 20 min after taking 10% AC diluted in 3% CMC, Duspatalin, TPLP 654 mg/kg/day, or TPLP

1962 mg/kg/day decreased the IM significantly, compared to the control lot. There was no difference between the lots given TPLP 654 mg/kg/day or TPLP 1962 mg/kg/day as compared to the Duspatalin lot. Forty minutes after taking AC, treatment with Duspatalin, TPLP 654 mg/kg/day, or TPLP 1962 mg/kg/day all decreased the IM, but the difference was not statistically significant ($P > .05$).

Effect of TPLP on the peristalsis and in vacuo smooth muscle tonus

The contractile frequency and amplitude of rabbit intestinal segments in different concentrations of TPLP luminal perfusion were recorded in organ bath model.

As shown in Table 4, TPLP 327 mg in 100 mL Tyrode's solution tended to reduce peristalsis frequency, but the difference was not statistically significant. TPLP at the concentrations of 654, 1308, and 1962 mg in 100 mL Tyrode's solution decreased peristalsis frequency significantly, compared to that of before using the drug.

The intestinal contractile amplitude was also recorded. With the increase of TPLP dose, there was a decrease of peristalsis movement in a concentration-dependent manner while Tyrode's solution did not affect peristalsis.

Effect of TPLP on the absorption of water and electrolytes from intestine into blood

The effect of TPLP on the absorption of water and electrolytes from intestine into blood was carried out using frog

TABLE 1. COLON MACROSCOPIC SCORES

Index	Macroscopic index				
	0	1	2	3	4
Colon weight (mg) (% increased/control)	<5	5–14	15–24	25–35	>35
Colon length (cm) (% decreased/control)	<5	5–14	15–24	25–35	>35
Stool score	Normal	Liquid/wet	No shape/stick	Diarrhea	
Inflammation	Normal	Mild inflammation, local edema	Average or widespread inflammation	Severe and spread infected inflammation	Ulcer or bleed

TABLE 2. COLON MICROSCOPIC SCORES

Index	Microscopic index			
	0	1	2	3
Epithelial damage	Normal	≤1/3 of tissue length	1/3–2/3 of tissue length	>2/3 of tissue length
Cellular infiltration	None	The degree of inflammatory cellular infiltration ≤1/3 of tissue length	The degree of inflammatory cellular infiltration 1/3–2/3 of tissue length	The degree of inflammatory cellular infiltration >2/3 of tissue length
Smooth muscle damage	Normal	≤1/3 of tissue length	1/3–2/3 of tissue length	>2/3 of tissue length

intestine *ex vivo* model. The put-in volume was constantly kept at 1 mL. After 60 min incubation, as shown in Table 5, the put-out fluid volumes were all reduced in every tested lot ($P < .001$). There was no difference in the fluid put-out volumes between Lot TPLP 654 mg/100 mL, Duspatalin 80 mg/100 mL, and the control ($P > .05$). Notably, TPLP 1962 mg/100 mL reduced the put-out fluid volume significantly when compared to the control ($P < .05$).

The electrolytes in the put-out fluid were analyzed by atomic absorption spectrometer. As shown in Table 6, Duspatalin 80 mg/100 mL decreased chloride concentration significantly in the put-out fluid compared to that of the control ($P < .05$). The drug also decreased sodium and potassium concentrations in the put-out fluid, but the difference is not significant when compared to the control. TPLP 654 mg/100 mL decreased sodium, potassium, and chloride concentrations in the put-out fluid but the difference was not statistically significant when compared to the control ($P > .05$). TPLP 1962 mg/100 mL significantly reduced chloride concentration compared to the control ($P < .05$). TPLP 1962 mg/100 mL also reduced sodium and potassium concentrations in the put-out fluid compared to the control, but the difference was not statistically significant ($P > .05$).

There was no difference between the put-out fluid volume of electrolytes (Na, K, Cl) in TPLP 654 mg/100 mL, 1962 mg/100 mL compared to Duspatalin ($P > .05$).

Effect of TPLP on MO-induced-IBS in mice

Animal overall health was monitored closely during the experiment. There is no visible difference between

TABLE 3. EFFECT OF TRANG PHUC LINH PLUS ON THE MOVEMENT OF ACTIVATED CARBON IN INTESTINAL MOTILITY

Lot	n	% of the intestinal length with AC per that from pylorus to cecum ($\bar{X} \pm SD$, %)	
		20 min	40 min
Group 1: control	9	71.80 ± 11.55	86.91 ± 10.69
Group 2: positive control (Duspatalin 80 mg/kg/day)	9	59.27 ± 11.94*	78.85 ± 12.56
Group 3: TPLP 654 mg/kg/day	9	61.71 ± 8.15*	78.35 ± 10.31 ^Δ
Group 4: TPLP 1962 mg/kg/day	9	56.89 ± 13.86*	77.06 ± 14.29

* $P < .05$ versus the control, ^Δ $P < .05$ versus the positive control.

AC, activated carbon; TPLP, Trang Phuc Linh Plus; SD, standard deviation.

all groups. As shown in Figure 2, all animal gained weight during the experimental course. There is no significant difference in body weights at the end of study between controls (A or B) and drug-treated IBS groups (C-1, C-2 and C-3). Animals in the IBS-induced group (C-0) were with significantly less weight than the control ($P < .05$).

Effect of drugs on the movement of AC in intestine

There was no difference in IM between the control and the solvent vehicle control (A vs. B) ($P > .05$). IM increased significantly in the model lots compared to the control and ethanol lots ($P < .05$). Duspatalin 80 mg/kg/day decreased IM clearly compared to the model ($P < .001$). TPLP 654 mg/kg/day, TPLP 1962 mg/kg/day reduced IM dramatically compared to the model, control, and solvent. There was no different effect of TPLP 654 mg/kg/day and Duspatalin 80 mg/kg/day on the IM ($P > .05$). TPLP 1962 mg/kg/day reduced IM statistical significantly compared to Duspatalin 80 mg/kg/day (Table 7).

TABLE 4. EFFECT OF TRANG PHUC LINH PLUS ON THE FREQUENCY AND AMPLITUDE OF PERISTALSIS AND *IN VACUO* SMOOTH MUSCLE TONUS

Lot	n	Before	After
Frequency (times/minute) ($\bar{X} \pm SD$)			
Lot 1: TPLP 327 mg/100 mL Tyrode's solution	6	8.83 ± 2.79	6.83 ± 2.93
Lot 2: TPLP 654 mg/100 mL Tyrode's solution	6	10.17 ± 1.17	7.5 ± 1.87*
Lot 3: TPLP 1308 mg/100 mL Tyrode's solution	6	10.67 ± 2.25	7.33 ± 1.86*
Lot 4: TPLP 1962 mg/100 mL Tyrode's solution	6	10.83 ± 1.6	6.50 ± 2.59**
Amplitude (mm) ($\bar{X} \pm SD$)			
Lot 1: TPLP 327 mg/100 mL Tyrode's solution	6	34.00 ± 6.75	15.67 ± 9.05**
Lot 2: TPLP 654 mg/100 mL Tyrode's solution	6	33.00 ± 12.12	14.50 ± 6.06***
Lot 3: TPLP 1308 mg/100 mL Tyrode's solution	6	32.33 ± 6.12	12.00 ± 8.41**
Lot 4: TPLP 1962 mg/100 mL Tyrode's solution	6	31.17 ± 9.87	10.17 ± 6.79**

* $P < .05$, ** $P < .01$, *** $P < .001$ versus before using the drug.

TABLE 5. EFFECT OF TRANG PHUC LINH PLUS ON THE ABSORPTION OF WATER FROM INTESTINE INTO BLOOD

Fluid volume (mL) Lot	n	Fluid volume put in (mL) ($\bar{X} \pm SD$)	Fluid volume put out (mL) ($\bar{X} \pm SD$)
Lot 1: control	8	1.00 $\pm$ 0.00	0.56 $\pm$ 0.16***
Lot 2: positive control (Duspatalin 80 mg/100 mL)	8	1.00 $\pm$ 0.00	0.53 $\pm$ 0.15***
Lot 3: TPLP 654 mg/100 mL	8	1.00 $\pm$ 0.00	0.51 $\pm$ 0.1***
Lot 4: TPLP 1710 mg/100mL	8	1.00 $\pm$ 0.00	0.39 $\pm$ 0.11 ^{Δ***##}

*** $P < .05$ versus the fluid volume put in the intestine, ^Δ $P < .05$ versus the control, [#] $P < .05$, ^{##} $P < .01$ versus the positive control.

Effect of TPLP on colon macroscopic and microscopic scores

After 10 days of using drugs, there was no significant difference between the colon macroscopic scores of the positive control (C-1) compared to negative controls (A and B) ($P > .05$). Colon macroscopic scores in IBS model (C-0) animals increased dramatically compared to group A and B ($P < .001$). There was no difference neither between TPLP treatment groups (C-2 and C-3) and controls (A and B groups), nor within TPLP groups. Scores regarding colon weight, length, stool, and inflammation were listed in Table 8.

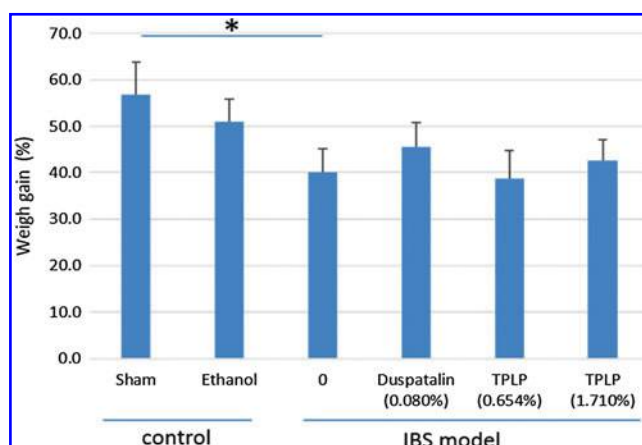
Effect of TPLP on colon histological damage induced by MO

Tissues were collected at 3 days post-MO application to determine effect of MO tissue damage in group A, B, and C-1. As shown in Fig. 3, there was no histological damage observed in the control and solvent (EtOH). Damaged epithelium, submucosal edema, loss of smooth muscle architecture, and a heavy cellular infiltration, or formed shallow ulcers were observed in the MO. Secretory glands increased actively and smooth muscles have mildly congestive blood vessels. At 7 days post-MO application, the structure of colon layers was clear. There were some inflammatory cells in secretory glands and interstitial tissues. Smooth muscles had mildly congestive blood vessels in the control and the

TABLE 6. EFFECT OF TRANG PHUC LINH PLUS ON THE ABSORPTION OF ELECTROLYTES FROM INTESTINE INTO BLOOD

Electrolytes (mmol/l) Lot	n	Na ($\bar{X} \pm SD$)	K ($\bar{X} \pm SD$)	Cl ($\bar{X} \pm SD$)
Lot 1: control	8	55.38 $\pm$ 26.99	10.88 $\pm$ 2.79	38.13 $\pm$ 9.08
Lot 2: positive control (Duspatalin 80 mg/100 mL)	8	35.13 $\pm$ 6.62	9.76 $\pm$ 1.40	27.88 $\pm$ 5.44 ^Δ
Lot 3: TPLP 654 mg/100 mL	8	40.38 $\pm$ 23.81	10.20 $\pm$ 2.95	29.38 $\pm$ 13.36
Lot 4: TPLP 1710 mg/100mL	8	47.38 $\pm$ 23.46	8.64 $\pm$ 4.17	22.88 $\pm$ 4.55 ^{ΔΔ}

^Δ $P < .05$, ^{ΔΔ} $P < .001$ versus the control.

FIG. 2. The percentage of animal body weight gain in MO induced IBS animals among different treatment groups. * $P < .05$.

solvent. Mucous surface was coated succus, glands increased to excrete and there were some inflammatory cells in stroma in the MO Lot.

At the end of the study, there was no difference about the histological damage score to the TPLPs Lot to that of the others (Fig. 4). There was no histological damage observed in the control, solvent (EtOH), model, and TPLPs lot.

DISCUSSION

IBS is characterized by chronic, recurrent abdominal pain and altered bowel habits and is currently defined by symptom criteria and the absence of detectable organic disease. The underlying pathophysiology remains incompletely understood. Despite considerable efforts by the scientific community and the pharmaceutical industry to develop novel pharmacological treatments aimed at chronic visceral pain, the traditional approaches to identify and evaluate novel drugs for this target have largely failed to translate into effective IBS treatments.¹¹

TABLE 7. EFFECT OF DRUGS ON THE MOVEMENT OF ACTIVATED CARBON IN INTESTINE

Lot	n	% of the intestinal length with AC per that from pylorus to cecum ($\bar{X} \pm SD$, %)
Lot 1: control	8	71.90 $\pm$ 10.94
Lot 2: solvent (Ethanol)	8	72.11 $\pm$ 9.47
Lot 3: model	8	83.81 $\pm$ 10.14 ^{*Δ}
Lot 4: positive control (Duspatalin 80 mg/kg/day)	8	57.37 $\pm$ 12.31 ^{*Δ ###}
Lot 5: TPLP 654 mg/kg/day	8	59.83 $\pm$ 12.11 ^{***Δ Δ ###}
Lot 6: TPLP 1962 mg/kg/day	8	42.78 $\pm$ 11.98 ^{***Δ Δ ###□}

Experiments were performed on day 17 or 10 days drug treatment on induced IBS animals. The AC travel distance was measured 20 min after AC administration.

* $P < .05$, ** $P < .01$, *** $P < .001$ versus the control, ^Δ $P < .05$, ^{ΔΔ} $P < .001$ versus the solvent, [#] $P < .05$, ^{###} $P < .001$ versus the model, [□] $P < .05$ versus the positive control.

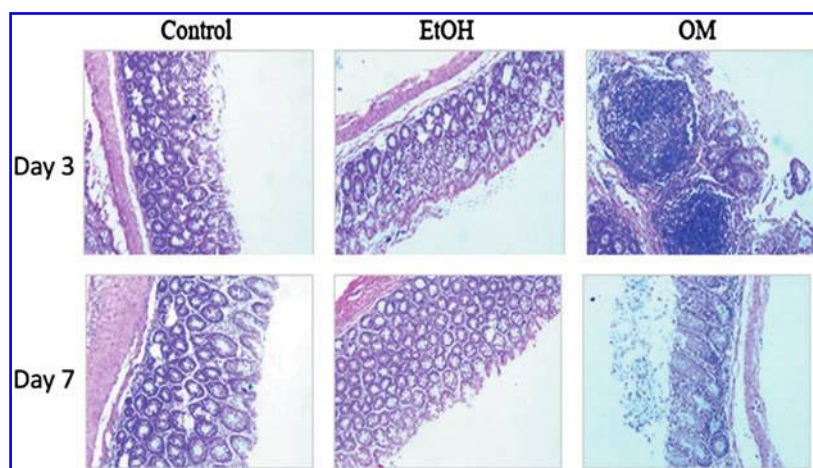


FIG. 3. The colon microscopic morphology (H&E) at 3 days and 7 days post MO application. Magnification: 250 $\times$. H&E, hematoxylin and eosin.

Current IBS treatment is based on an array of dietary or medicinal, psychotherapeutic, and alternative measures. TPLP is a food supplement that has been approved in Vietnam as a supportive treatment for IBS and other gastrointestinal disorders.

The study of Kimball *et al.* showed that acute colitis induction by MO resulted in later development of IBS-like conditions in mice.⁹ Mice given a single administration of MO developed a severe colitis that peaked at day 3, was reduced at day 7, and was absent by day 14. At the peak response, there was body weight loss, colon shrinkage, thickening and weight increases, distension of the proximal colon, and diarrhea. Macroscopic inspection of the distal colon revealed a discontinuous pattern of inflammatory damage and occasional transmural ulceration. Histological examination showed loss of epithelium, an inflammatory infiltrate, destruction of mucosal architecture, edema, and loss of circular smooth muscle architecture. MO administration also increased transit of a carmine dye bolus from 58% of the total length of the upper GI tract in untreated age-matched controls to as high as 74% when tested at day 28 post-MO. MO induced a rapid, acute, and transient colitis and, in the longer term, functional changes in motility that are observed when there was no gross inflammation.

Wagh *et al.* used this model to evaluate both the post-treatment and pretreatment of *Lactobacillus plantarum* 299v

on IBS.¹² MO was administered rectally to induce IBS. The study examined the effect of *L. plantarum* 299 v on colonic inflammation and motility. *L. plantarum* 299 v (1×10^9 cfu) was gavaged for up to 28 days, beginning either 7 days before (pretreatment) or 8 days after oil of mustard administration (post-treatment). Small intestinal transit was assessed via the dye-transit score technique at day 20. *L. plantarum* 299 v reduced inflammation at both day 4 and 20 and normalized intestinal transit rates in the oil of mustard murine IBS model.

MO-induced IBS in mice is considered a model of functional disorders that mimics PI-IBS in humans.¹¹ For this reason we used this model to evaluate the treatment of TPLP on IBS.

Our results indicated that TPLP 654 mg/kg/day and TPLP 1962 mg/kg/day have therapeutic and preventive effects on IBS induced by MO. TPLP decreased IM significantly compared to the control and similar to Duspatalin 80 mg/kg/day. IM effect of TPLP 1962 mg/kg/day was better than Duspatalin 80 mg/kg/day in the treatment study. TPLP 654 and 1962 mg/kg/day reduced the IM *in vivo* and *in vitro*. The compound decreased the fluid volume and electrolytes in intestine by increasing the absorption of water and electrolytes from intestine into blood.

Our study proved the combination of probiotic-derived biological agent and herbal ingredients that have been

TABLE 8. EFFECT OF DRUGS ON COLON MACROSCOPIC AND MICROSCOPIC INDEXES

Lot	n	Colon macroscopic scores ($\pm$ SD)	Colon microscopic indexes ($\pm$ SD)			
			Colon weight (mg)	Colon length (cm)	Stool score	Inflammation
A: control	8	0.13 $\pm$ 0.35	554.25 $\pm$ 128.29	83.75 $\pm$ 14.08	0.13 $\pm$ 0.35	0.00 $\pm$ 0.00
B: solvent (Ethanol)	8	0.25 $\pm$ 0.46	549.50 $\pm$ 98.95	90.00 $\pm$ 14.14	0.25 $\pm$ 0.46	0.00 $\pm$ 0.00
C-0: model	8	2.5 $\pm$ 0.93*** Δ Δ	565.88 $\pm$ 57.45	86.25 $\pm$ 11.88	2.25 $\pm$ 0.71*** $\Delta\Delta\Delta$	0.25 $\pm$ 0.46
C-1: positive control (Duspatalin 80 mg/kg/day)	8	0.63 $\pm$ 0.74###	560.38 $\pm$ 51.98	91.25 $\pm$ 15.53	0.25 $\pm$ 0.46###	0.38 $\pm$ 0.52
C-2: TPLP 654 mg/kg/day	8	0.50 $\pm$ 0.85###	560.50 $\pm$ 92.85	85.00 $\pm$ 15.81	0.30 $\pm$ 0.48###	0.20 $\pm$ 0.42
C-3: TPLP 1962 mg/kg/day	8	0.44 $\pm$ 0.88###	571.33 $\pm$ 128.17	87.78 $\pm$ 9.72	0.22 $\pm$ 0.44###	0.22 $\pm$ 0.44

* $P < .05$, ** $P < .01$, *** $P < .001$ versus the control, $\Delta P < .05$, $\Delta\Delta P < .01$, $\Delta\Delta\Delta P < .001$ versus the solvent, # $P < .05$, ## $P < .01$, ### $P < .001$ versus the model.

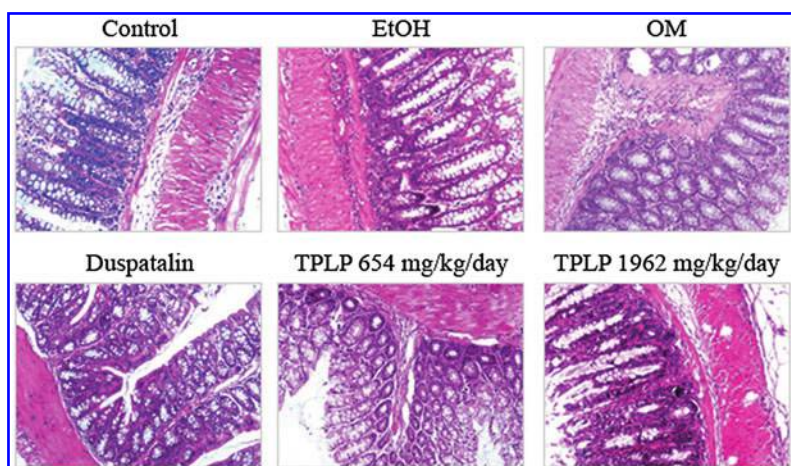


FIG. 4. The colon microscopic morphology (H&E) at the end of the treatment studies. The structure of colon layer was clear in all lots. There were some inflammatory cells in secretory glands and interstitial tissues. Smooth muscles had mildly congestive blood vessels. Mucous surface was coated succus and with increased glands. There were more inflammatory cells in stoma.

traditionally used for management of gastrointestinal diseases in Vietnam have a remarkable therapeutic effect *in vitro* and *in vivo* in IBS animal models. Given the facts that the ingredients of the TPLP are readily available, less expensive, nontoxic, and the satisfactory empirical clinical efficacy for patients with IBS and other inflammatory bowel diseases, future clinical trial to determine the therapeutic value of the compound is a desirable direction in exploring the new treatment options for IBS.

CONCLUSION

Our study demonstrated that TPLP had good effect on IBS animal models and *ex vivo* models. It might explain the positive outcome of its empirical application as nutritional therapeutic products for management of IBS and inflammatory bowel diseases.

AUTHOR DISCLOSURE STATEMENT

No competing financial interests exist.

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